

Stakeholder Engagement Plan

Optimising RAG Document Retrieval for Agronomic Advice

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Stakeholder Group 1: Knowledge-Seeking Farmers (Long-Term Planning)

Who they are: Smallholder farmers planning new crops or improving existing practices, without an immediate crisis driving the need. Many rely on trial-and-error learning, peer advice, or outdated information, with limited access to formal extension support.

What matters most to them: Reliable, trustworthy information they can act on with confidence. They value guidance that prevents costly mistakes, is accessible through phones, is understandable without technical jargon, and is validated by results other farmers have seen, since trust is built through consistency, not a single interaction.

Potential Benefits:

- Reduced crop losses through access to correct advice before investing in seeds or inputs
- Avoided waste on the wrong fertiliser or treatment based on a misdiagnosed issue
- A faster learning curve, building confidence over weeks rather than seasons of trial and error

Potential Harms/Concerns:

- If retrieval is uneven across regions (our data shows Nigeria has 133 of 695 documents while countries like Togo have only 3), farmers in underrepresented areas may get weaker guidance without realizing why
- Risk of misapplying advice meant for a different climate, soil type, or region to their own context
- Loss of trust in the system if even a single irrelevant result is returned early on

Ethical/Social Considerations: Since 637 of our 695 documents are synthetically generated, there's a risk farmers could treat this advice as fully expert-verified when its real-world grounding hasn't been independently confirmed.

Stakeholder Group 2: Farmers Facing an Active Crop Emergency

Who they are: Farmers dealing with a sudden pest outbreak, disease, or weather damage requiring an immediate, specific answer, often under real time pressure and with a crop actively at risk.

What matters most to them: Speed and specificity above all else. They need a solution to their exact problem, not general information, and need confidence the guidance will work before acting on it, especially if a fix is costly or risky to attempt.

Potential Benefits:

- Crisis mitigation, saving a portion of a crop that might otherwise be lost entirely
- Reduced panic and better decision-making under stress, replacing guesswork with a validated starting point

Potential Harms/Concerns:

- A near miss document, one that looks related but addresses a different underlying issue, could lead to a wrong, costly decision made under time pressure
- Incomplete retrieved information could leave a farmer only partway through a needed fix
- Overreliance on the system's output without the involvement of a human expert during a high-stakes decision

Ethical/Social Considerations: Urgency increases the risk of treating retrieved documents as a final answer rather than supporting evidence, which is exactly why our project deliberately limits its scope to retrieval rather than generating standalone advice.

Engagement Method 1: Distributed Dialogue (Group 1)

Why this method fits: Distributed Dialogue is a decentralized approach where engagement happens across multiple informal settings and times rather than one central event, well suited to farmers spread across regions with limited simultaneous availability. Rather than requiring travel to a workshop, this method meets farmers where they already are, through phone conversations, community radio, and peer networks, and allows feedback to happen in farmers' own languages rather than only formal English.

How it would be conducted: Periodic, low-pressure conversations with a small group of farmers across a few regions, gathered through phone calls or existing community channels, focused on which topics feel underserved and whether retrieved advice matches how they naturally phrase their questions. Feedback would be gathered iteratively over time rather than as a single one-off session, allowing the team to identify patterns rather than isolated opinions.

Engagement Method 2: Participatory Design (Group 2)

Why this method fits: Crisis farmers hold direct, lived expertise in what a real agricultural emergency actually demands, information a researcher cannot fully anticipate secondhand. Participatory Design involves these farmers as active co-designers rather than passive testers, capturing that expertise directly in how an emergency-focused version of the system should work.

How it would be conducted: A small group of farmers who have previously experienced crop emergencies would be invited to help shape decisions such as whether speed matters more than exhaustive detail, what information should surface first, and whether a voice-based or simplified format better serves someone in the middle of a crisis rather than reading a full document.

Influence on Project Design

- **Dataset:** Both groups' feedback would help identify coverage gaps, crops, countries, or problem types underrepresented or missing entirely, directly informing which documents to prioritise adding.
- **Model design:** Crisis farmer input could reveal that urgent queries need different retrieval weighting (favouring speed and directness) than exploratory planning queries, potentially requiring separate tuning for each use case.
- **Outputs:** Feedback on language and phrasing could shift how results are presented, for example, offering a simplified summary alongside the full retrieved document rather than only the raw factsheet.
- **Evaluation:** Real farmer queries and outcomes would supplement our current nDCG@5 evaluation, testing whether strong performance on the provided training data holds up against how real users actually phrase their problems.
- **Deployment decisions:** Findings from both engagement methods would inform whether any future deployment needs a human intermediary, such as an extension worker relaying results, rather than farmers accessing raw retrieval output directly, particularly for the crisis use case where the cost of an error is highest.
- **Dataset:** Extension worker feedback would add a professional verification layer, flagging which retrieved documents are inaccurate or outdated based on their field expertise, not just which ones are irrelevant.
- **Deployment decisions:** Extension worker input would help determine whether the tool is best positioned as a direct-to-farmer resource or as a support tool specifically for intermediaries like themselves, given their concerns about professional credibility and overreliance.

Stakeholder Group 3: Agricultural Extension Workers

Who they are: Government or NGO-affiliated agricultural advisers who work directly with farmers, offering guidance, diagnosing crop issues, and relaying expert knowledge. They are often stretched thin, covering large numbers of farmers with limited time per visit, and act as a trusted intermediary between formal agricultural knowledge and farming communities.

What matters most to them: Tools that make their limited time more effective, rather than adding extra work. They value being able to quickly verify or supplement their own knowledge when facing an unfamiliar case, and they value maintaining the trust farmers place in them, meaning they need confidence in a tool before relying on it in front of a farmer.

Potential Benefits:

Faster access to relevant reference material when facing a crop issue outside their immediate expertise, without needing to consult external resources manually

Support for covering more farmers effectively, since quick retrieval could reduce time spent searching for the right guidance during a farm visit

A shared, consistent reference point across a team of extension workers, reducing inconsistent advice between different officers

Potential Harms/Concerns:

If the system's retrieval quality is uneven across regions or crops, an extension worker could unintentionally pass along a mismatched near miss document as if it were reliable

Overreliance risk, if used carelessly, the tool could replace rather than support an extension worker's own judgement and field experience

Professional credibility risk, if the tool gives incorrect guidance in front of a farmer, it could damage the extension worker's standing, not just the tool's

Ethical/Social Considerations: Since extension workers are trusted intermediaries, any error the tool introduces doesn't stay contained, it can propagate through their relationship with the farmers who depend on them, making accuracy and clear communication of the tool's limitations especially important for this group.

Engagement Method 3: User Committees (Group 3)

Why this method fits: Unlike farmers who are dispersed and often engage informally, extension workers are a smaller, more defined professional group with existing structures, such as regional offices or professional networks, making a standing committee realistic rather than logistically difficult. A recurring committee also matches how extension workers already engage with new tools and guidance in a professional capacity, through structured review rather than ad hoc conversation.

How it would be conducted: A small committee of extension workers, drawn from a few different regions to capture varied field conditions, would meet periodically to review system outputs against real cases they've encountered, flag documents that seem unreliable or poorly matched, and discuss whether the tool genuinely saves them time or adds friction to their existing workflow. Their structured, recurring feedback would carry more domain authority than farmer feedback alone, given their professional training.